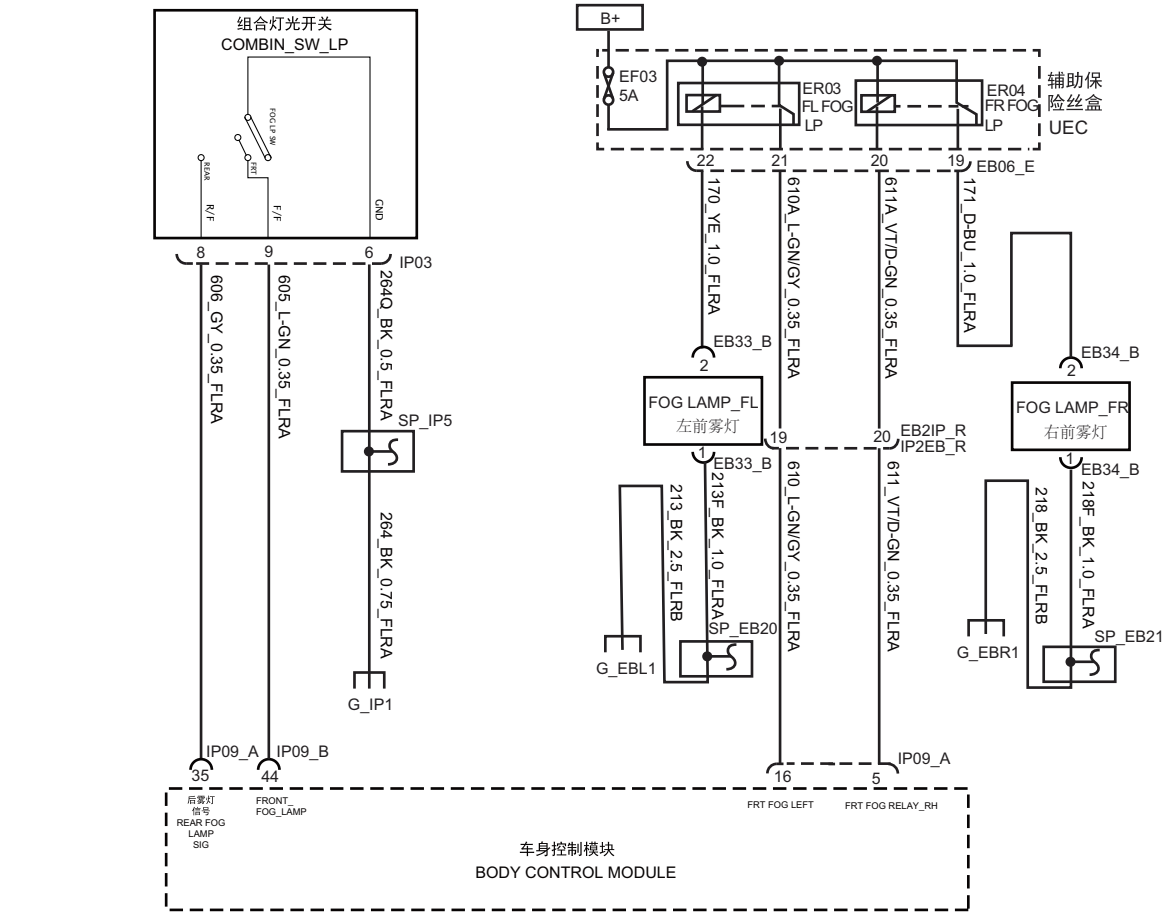
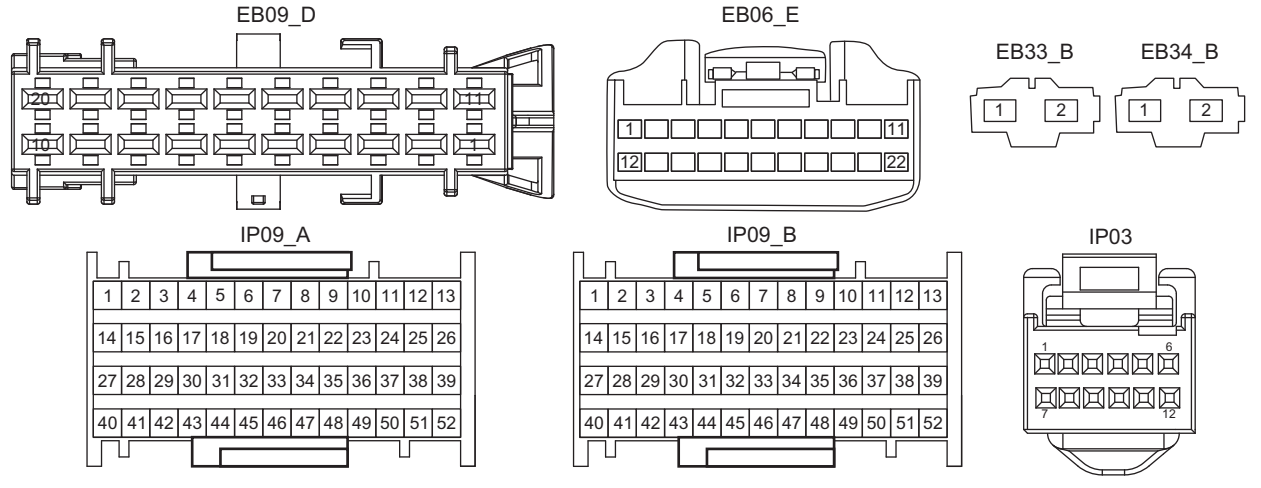
## B1250-12、B1252-11、B1250-13、B1251-11、B1251-12、B1251-13、U0140-87、B1252-12、B1252-13

| **Fault diagnosis code** |
| --- |
| B1250-11：Front fog light circuit short to ground |
| B1250-12：Front fog light circuit short to battery |
| B1250-13：Front fog light circuit open |
| B1251-11：Front fog light circuit short to ground |
| B1251-12：Left front fog light circuit short to battery |
| B1251-13：Left front fog light circuit open |
| B1252-11：Right front fog light circuit short to ground |
| B1252-12：Right front fog light circuit short to battery |
| B1252-13：Right front fog light circuit open |
| **Detection tools** |
| 1 Multimeter and diagnostic apparatus |
| **Possible causes** |
| • Circuit,terminal and adapting piece fault<br>• BCM fault<br>• Left rear front fog lamp circuit fault<br>• Insurance fuse fault<br>• Relay fault |
| **Conditions for fault code operation** |
| Turn the ignition switch to ON position. |
| **Conditions for fault code setting** |
| B1250-11 |
| Front fog lamp output circuit short to ground |
| B1250-12 |
| Front fog lamp output circuit short to battery |
| B1250-13 |
| Front fog lamp output circuit open |
| B1251-11 |
| Front fog lamp output circuit short to ground |
| B1251-12 |
| Front fog lamp output circuit short to battery |
| B1251-13 |
| Front fog lamp output circuit open |
| B1252-11 |

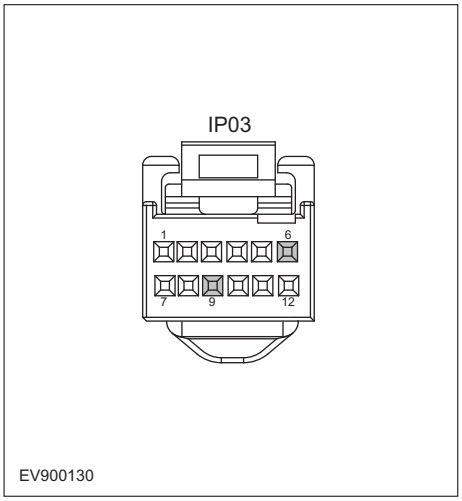
| |
|---|
| Front fog lamp output circuit short to ground |
| B1252-12 |
| Front fog lamp output circuit short to battery |
| B1252-13 |
| Front fog lamp output circuit open |
| **To set the effect of a fault code condition** |
| Front fog lamp function fails to effect |
| **Description of circuit diagram** |
| |

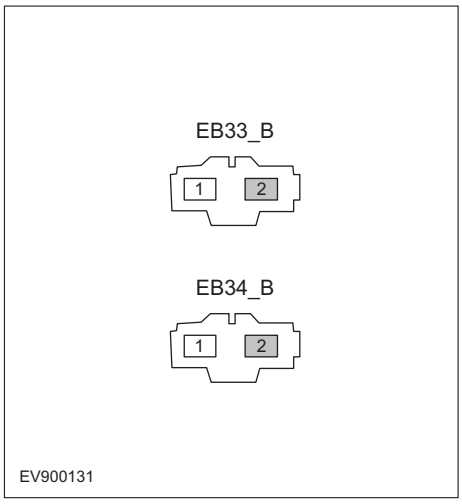
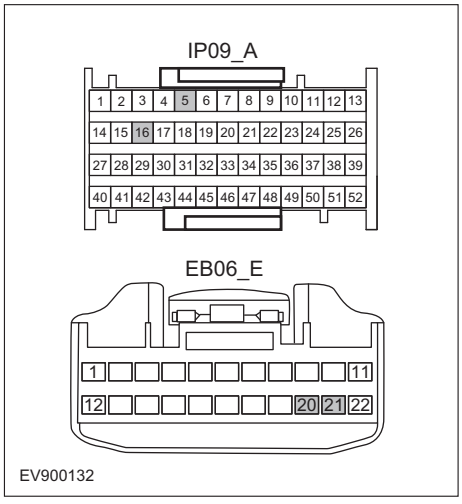
**Circuit diagram**



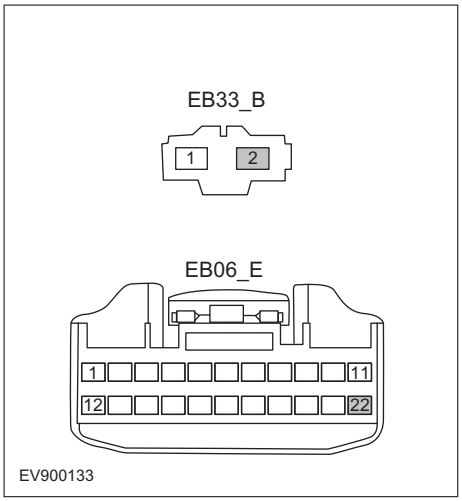



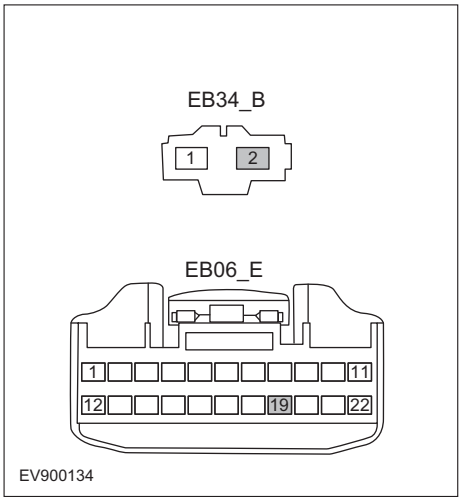

EV900129

| Troubleshooting steps | | |
|---|---|---|
| Step 1 | General inspection. | |
| | | 1 Check the fuse EF03 for damage.<br>2 Check the relay ER03/ER04 for damage.<br>3 Check battery capacity.<br>4 Check the harness plug of air conditioning control module for damage, poor contact, aging and looseness.<br>5 Check whether the result is normal or not? |
| | | **NO →** Repair or replace the faulted parts. |
| **YES ↓** | | |
| Step 2 | Check fault codes. | |
| | | 1 Connect the fault diagnosis instrument to the diagnosis interface.<br>2 Turn the ignition switch to ON position.<br>3 Check the body control module with a diagnostic apparatus.<br>4 If there is any fault code except for B1250-11、B1250-12、B1250-13、B1251-11、B1251-12、B1251-13、B1252-11、B1252-12、 B1252-13? |
| | | **YES →** Refer to:DTC Summary list(BCM). |
| **NO ↓** | | |
| Step 3 | Check the combination lights switch (fog lamp switch)(parts). | |
|  | | 1 Turn the ignition switch to OFF.<br>2 Place the combination light switch in the “front fog lamp”position and measure the resistance between the combination light switch terminals(combination switch).<br>Measuring circuit: resistance between terminal 9 on plug IP03 and terminal 6 on plug IP03.<br>**Standard value: ≈ 0Ω**<br>3 Check whether the result is normal or not? |
| | | **NO →** Replace combination lights. |
| **YES ↓** | | |

| Step 4 | Check the supply power of front fog lamp when it is started or operating. |
|---|---|
|  | 1 Turn the ignition switch to OFF.<br>2 Disconnect the harness plug EB33_B of left front fog lamp ;<br>3 Disconnect the harness plug EB34_B of right front fog lamp.<br>4 Turn the ignition switch to ON.<br>5 Place the combination light switch in the “front fog lamp”position.<br>6 Measure the voltage between the harness terminal of front fog lamp and ground.<br>Measuring circuit: voltage between terminal 2 on plug EB33_B and ground.<br>Measuring circuit: voltage between terminal 2 on plug EB34_B and ground.<br>**Standard value:10~14V**<br>7 Check whether the result is normal or not? |
| | **NO →** Repair or replace the faulted parts. |
| **YES ↓** | |
| Step 5 | Check the circuit between relay of front fog lamp and BCM (open circuit). |
|  | 1 Turn the ignition switch to OFF.<br>2 Disconnect the harness plug IP09_A of BCM.<br>3 Disconnect the harness plug EB06_E of UEC.<br>4 Measure the resistance between the harness terminal of UEC and the BCM.<br>Measuring circuit: resistance between terminal 20 on plug EB06_E and terminal 5 on plug IP09_A.<br>Measuring circuit: resistance between terminal 21 on plug EB06_E and terminal 16 on plug IP09_A.<br>**Standard value: ≈ 0Ω**<br>5 Check whether the result is normal or not? |
| | **NO →** Repair or replace the faulted parts. |
| **YES ↓** | |

| Step 6 | Check the circuit between relay of front fog lamp and front fog lamp (open circuit). |
|---|---|




1 Measure the resistance between the harness terminal of UEC and front fog lamp.

Measuring circuit: resistance between terminal 22 on plug EB06_E and terminal 2 on plug EB33_B.

Measuring circuit: resistance between terminal 19 on plug EB06_E and terminal 2 on plug EB34_B.

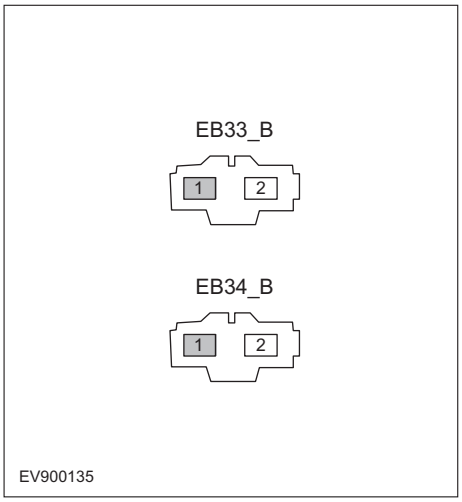
**Standard value: ≈ 0Ω**

2 Check whether the result is normal or not?

**NO →** Repair or replace the faulted parts.

**YES ↓**

| Step 7 | Test whether the grounding circuit of front fog lamp is open. |
|---|---|




1 Turn the ignition switch to OFF.

2 Measure the resistance between the harness terminal of front fog lamp and ground.

Measuring circuit: resistance between terminal 1 on plug EB33_B and ground.

Measuring circuit: resistance between terminal 1 on plug EB34_B and ground.

**Standard value: < 1Ω**

3 Test whether the resistance is less than 1Ω?

**NO →** Repair or replace the faulted parts.

<table>
<tr><td colspan="3">YES ↓</td></tr>
<tr><td>Step 8</td><td colspan="2">Test whether BCM is norm.</td></tr>
<tr><td colspan="2"></td><td>1 Turn the ignition switch to OFF.<br>2 Disconnect and check the harness plug EB09_C、EB09_D、IP09_A、IP09_B of BCM.<br>3 Check and repair.<br>• Corrosion(install new adapting piece or terminal-cleaning module and pin).<br>• Pin damaged or bent-install new terminal/pin.<br>• Fly out of pins-install new pins as needed.<br>4 Reconnect the BCM plug and all other previously disconnected plug. Make sure the pin is properly positioned and fixed.<br>5 Run the system and determine if the problem persists?</td></tr>
<tr><td colspan="2"></td><td>YES → Replace the BCM.</td></tr>
<tr><td colspan="3">NO ↓</td></tr>
<tr><td colspan="3">At this point the system is functioning normally. The problem may be caused by module connections. Address the root cause of any connector or pin problems,</td></tr>
<tr><td colspan="3">Next Step ↓</td></tr>
<tr><td colspan="3">Diagnosis end.</td></tr>
</table>